

Retail Sentiment Intelligence

Real-Time Social Media Mining and Trust-Aware Sentiment Analysis
Using Large Language Models for Retail Feedback Optimization

BITS ZG628T · Dissertation · Post Mid-Semester Presentation

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Agenda

Focus of today — post mid-semester work only

1

Since Mid-Sem — What's New

Post-midsem workstream map

2

Review & Validate (HITL)

Human-in-the-loop correction workflow

3

Smart Reply — Triple Draft

GPT-4o + Mistral 7B + Smart Composer

4

Smart Reply — Prompt & Few-Shot

Prompt template + past validated replies

5

Smart Reply — Worked Example

One complaint → three drafts

6

Learning Loop

Corrections + posted replies → retraining

7

Post Explorer

Multi-facet search across analysed posts

8

Post Lifecycle (Kanban)

Triage → Acknowledged → In-Progress → Resolved

9

Insights & Competitor Analysis

Strategic weekly summaries + cross-brand

10

Notification Centre

In-app + email + Slack, group-routed

11

ModernBERT — Final Results

Macro-F1 0.6272 → 0.7642 (+13.7 pts)

12

Vision — Multi-Pass Payoff

Hallucination 50% → 0%, extraction 25% → 75%

13

Trust-Score Evaluation

15% flagged; 12 of 15 confirmed by annotator

14

Live Demo

Dashboard walkthrough (screenshots)

15

Conclusions & Future Work

RQ1–RQ4 outcomes + realistic roadmap

Since Mid-Sem — What's New

Mid-sem shipped the pipeline; post-midsem made it usable for analysts

Delivered at mid-sem

- 6-layer offline-first pipeline (25 subreddits)
- ModernBERT fine-tuning up to Stage 2 (F1 0.7285)
- Multi-pass vision with 0% hallucination on pilot
- Trust-score design + weighted formula
- Brand-Health + Alert Feed dashboard pages
- Group-routed email / Slack notifications

Added post-midsem (this deck)

- Review & Validate — human-in-the-loop UI + backend
- Smart Reply Composer — dual-draft (rule + FLAN-T5)
- Learning-loop store — corrections & posted replies
- Post Explorer — multi-facet search
- Post Lifecycle — Kanban with 2-step Resolve
- Insights & Competitor Analysis page
- In-app Notification Centre mirroring the digest
- ModernBERT Stage 3 (final) — Macro-F1 0.7285 → 0.7642 (5-fold OOF)
- Trust-score end-to-end evaluation (n = 200)
- SQLite schema mirroring Cosmos DB partitioning

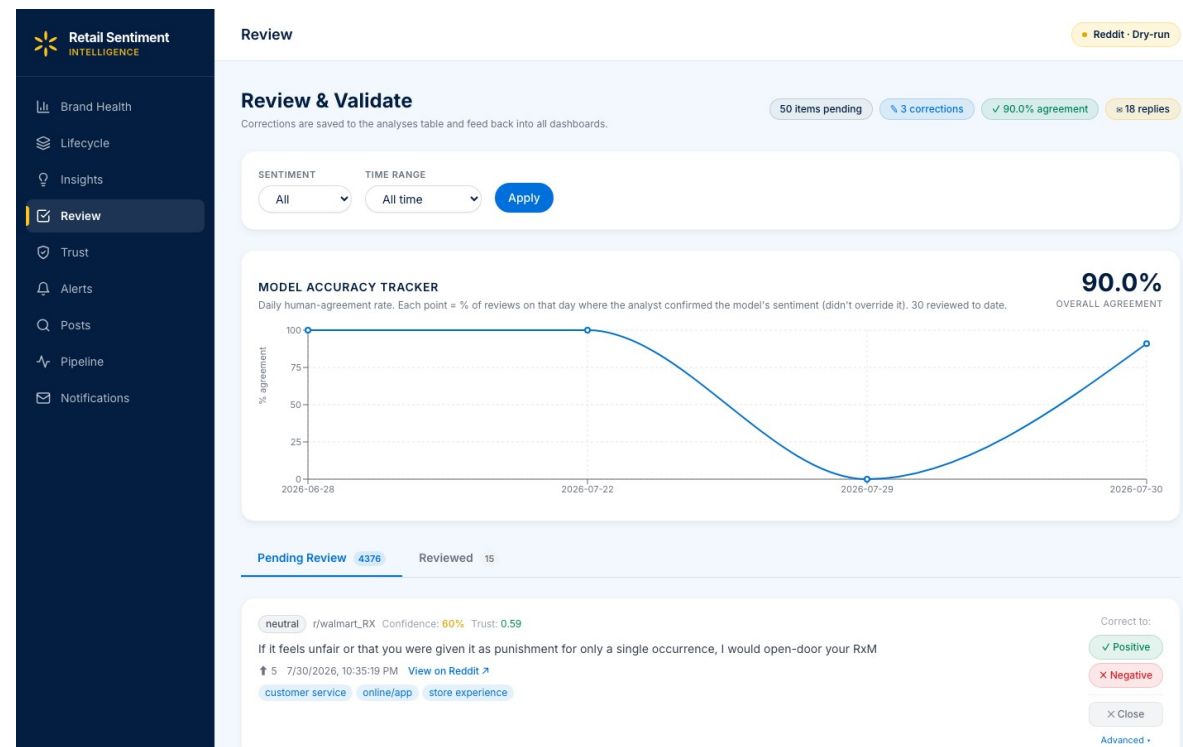
Framing

Mid-sem answered can the pipeline produce trust-weighted sentiment? Post-midsem answers can an analyst act on it, correct it, and feed the corrections back into the model?

Review & Validate — Human-in-the-Loop

Every prediction can be corrected; every correction is logged

- Queue sorted by priority (P1 first) · trust × confidence
- Analyst can override sentiment and aspect tags in one click
- Corrections written to feedback table with analyst + timestamp
- Generate Drafts → three reply options (GPT-4o + Mistral + Smart Composer)
- Analyst picks A, B or C, edits inline, and posts to Reddit
- Posted replies also captured → future few-shot pool



Why it matters

Turns a passive dashboard into a training signal — corrections harvested here drive the ModernBERT re-training loop and reply few-shot pool.

Smart Reply Composer — Triple Draft

One prompt, three generators; analyst picks the best draft

Draft A

GPT-4o

Walmart LLM Gateway

Strong reasoning, safest tone
Guardrails via Walmart proxy
~\$0.0002 / reply
Fallback: direct OpenAI

Draft B

Mistral 7B-Instruct

Local Ollama (:11434)

Open-weights, free
Runs fully offline
~15 s warm latency
Strong at retail jargon

Draft C

Smart Composer

Deterministic template

No LLM, always available
Zero latency, zero cost
Content-aware phrase pools
Safety-net fallback

Fallback logic — no draft is ever missing

- GPT-4o unavailable (no gateway key / consumer-ID / network) → Smart Composer draft in slot A
- Mistral unavailable (Ollama not running) → Smart Composer draft in slot B
- Smart Composer always produces slot C — guaranteed reply
- UI labels each card with the actual model used, plus an offline-fallback badge when applicable

Smart Reply — Prompt Design & Few-Shot

Same prompt fed to GPT and Mistral; Smart Composer skips the LLM step

Prompt template (built in `_build_reply_prompt`)

You are a senior Walmart customer-care analyst replying on Reddit.
Write ONE reply to the customer below. Keep it 2-4 sentences,
empathetic, specific to their complaint, no corporate jargon,
no hashtags, no emojis. Do NOT promise refunds you can't verify;
invite them to DM order details if action is needed. Sign off as
a real person, not a brand.

Example customer post: {past_post_1}
Example analyst reply: {past_reply_1}
Example customer post: {past_post_2}
Example analyst reply: {past_reply_2}

Subreddit: r/{subreddit}
Customer ({author}) complaint about: {aspect_1}, {aspect_2}
Customer post:
{post_title + post_body, ≤ 1200 chars}

Reply:

Few-shot source — feedback table

- Top-3 past validated replies for the same aspect
- Written by human analysts, already posted to Reddit
- Refreshed on every Generate click → adapts over time

Style guardrails baked into the prompt

- 2 – 4 sentences, empathetic, specific
- No corporate jargon, no hashtags, no emojis
- No unverifiable refund promises → DM to continue
- Sign off as a person, not a brand

Smart Reply — Worked Example (all three drafts real, live-captured)

raw_posts.id = reddit_1u2bgdw · r/samsclub · gateway ✓ · Ollama ✓ · Smart Composer ✓

CUSTOMER COMPLAINT · r/samsclub

Why do you guys sell whole pizzas made hours ago to customers?

"The very few times I've gotten a whole pie, it'll be stuff premade and left in the hot case for like an hour 30mins before it's in my hand. How can I tell? They put a sticker with the date and the pizza looks and taste hours old. Pizza is meant to be made to order and waited for."

ModernBERT: negative (conf 0.9999997) · aspects: customer service · product quality · store experience · trust 0.66

 [Open original on Reddit](#)

A · GPT-4o (Walmart Gateway)

Hi u/there, I'm sorry to hear about your experience with the pizzas. I understand how disappointing it must be to get something that feels less fresh than expected. While the hot case is meant to provide convenience, I agree that quality should never be sacrificed. If you'd like, feel free to DM me with the details of your store location so we can look into this and address it properly. – [Your Name]

B · Mistral 7B (local Ollama)

Hi u/there, I appreciate your concern about the quality of our prepared pizzas. It's important to us that our customers receive fresh and delicious food. The stickers with dates are there to help us manage the rotation of our products, ensuring they are consumed before they become stale. If you ever encounter a pizza that doesn't meet your expectations, please feel free to DM me the details of your order so I can look into it further. Thank you for taking the time to share your feedback.

C · Smart Composer (no-LLM)

Hi u/there — thanks for flagging this. completely understand the frustration around your order. Drop us a private message with the order # or store info and we'll take it from here. — Walmart Care

One prompt fed to all three engines. Analyst picks the best draft, edits inline, clicks Save & Open Reddit → the posted reply is written to feedback and becomes the next post's top few-shot example.

Learning Loop — Feedback → Retraining

Every human action becomes a training signal — hot loop (live) + warm loop (monthly)

Label / aspect correction	Trust-score override	Posted reply	Lifecycle transition
Review & Validate	Review & Validate	Smart Reply Composer	Kanban board

feedback table (SQLite / Cosmos DB) — one row per human action, JSON payload, partition_key = analyst_id

HOT LOOP • in-context, live

- Every Generate Drafts click queries feedback
- SELECT ... WHERE kind = 'auto_reply_posted'
- ORDER BY created_at DESC LIMIT 5 → top 3 pairs
- Injected into the prompt as few-shot examples
- Effect visible on the next reply — no training

WARM LOOP • supervised, monthly

- Export corrections since last train date
- Append to Walmart-200 → Walmart-N (augmented)
- Rerun ModernBERT Stage-3 with 5-fold OOF CV
- New checkpoint wins only if OOF F1 improves
- Symlink flipped in config/models.yaml

Two failure modes eliminated by design

(1) hot loop uses in-context learning — no retrain, no catastrophic forgetting. (2) warm loop uses 5-fold OOF CV — no leakage between train and eval sets.

Post Explorer — Multi-Facet Search

Find the exact 20 posts an analyst needs to look at today

Facets

- Sentiment · neg / neu / pos
- Confidence slider (0.50 – 1.00)
- Trust-score slider (0.00 – 1.00)
- Subreddit — multi-select from 25 tracked
- Aspect — 8-item retail taxonomy
- Date range — today / week / month / custom
- Full-text search — title + body



Brand Health

Lifecycle

Insights

Review

Trust

Alerts

Posts

Pipeline

Notifications

Posts

Reddit · Dry-run

Post Explorer

Search the analyzed-post archive by subreddit, sentiment, aspect or trust score.

SUBREDDIT

SENTIMENT

ASPECT

RANGE

MIN TRUST

LIMIT

Search

e.g. walmart

All

e.g. pricing

All

0.0-1.0

50

Showing 50 of 83,006 matches (raise the Limit to see more)

neutral

r/WalmartEmployees

↑ 1

7/31/2026, 3:33:40 PM

trust 0.55

onepay

newbie here. how does instapay work on the onepay app?

View on Reddit

negative

r/AmazonPrime

↑ 1

7/31/2026, 3:33:25 PM

trust 0.66

Amazon promised me a \$150 shipping refund in chat to make me return an item, then refused to pay. Can they just lie like that?

Hey everyone, I need some advice or to know if anyone else has dealt with this nightmare. I recently had to return an item to Amazon internationally. When I was setting up the return through live chat, the support agent explicitly told me: "You can use any carrier to return the item. You need to pay to return the item. But the good thing is, we can help you with the refund for return shipping charges up to \$150." Because of that exact, written promise, I went ahead, paid out of pocket, and shipped the...

View on Reddit

negative

r/eCommerce

↑ 1

7/31/2026, 3:27:20 PM

trust 0.66

Struggling with cancelled parcels in fulfillment

We've been seeing a rise in cancelled orders on TikTok, Lazada, and Shopee that are already processed and packed in our fulfillment area. The problem is, by the time we realize they're cancelled, the items are fully packed— meaning wasted packaging materials and extra work. I'm trying to figure out how to catch these cancellations earlier in the workflow so we can save on cartons, fillers, and waybills. Has anyone here dealt with this? How do you identify cancelled items before they even reach...

View on Reddit

positive

r/CustomerService

↑ 1

7/31/2026, 3:24:38 PM

trust 0.66

Dear ppl who call in to customer service

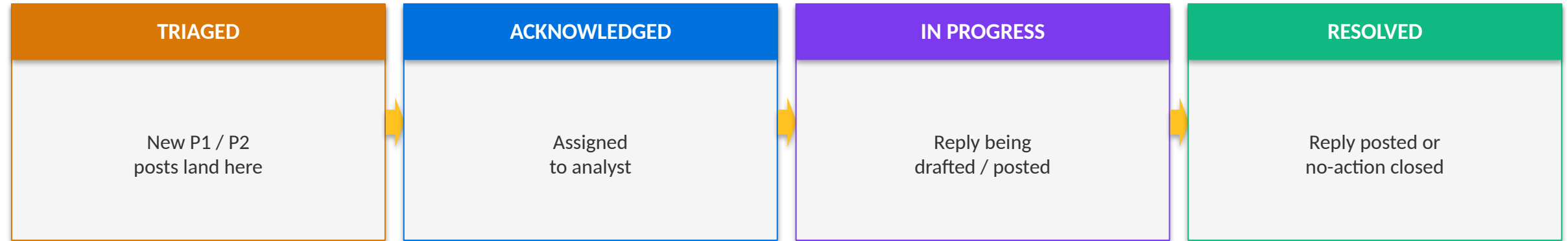
I am a call center agent, I actually enjoy my job! but for those who call into anywhere yelling and fussing about goofy shyt. or a package coming a day late, or ur broke ass begging for a discount... get it together 🤔🤔 I hope you get hung up on until you know how to talk to people. I hang up on customers all the time🤔 soon as

Per-post card

Title + body excerpt · sentiment badge with confidence % · trust tier (H / M / L) · aspect chips · subreddit + timestamp · actions: Review · Add to Lifecycle · View Details.

Post Lifecycle — Kanban Workflow

Track a complaint from Triage to Resolved with SLA visibility



The screenshot shows the 'Post Lifecycle' interface in the Retail Sentiment Intelligence tool. The left sidebar contains navigation options: Brand Health, Lifecycle (selected), Insights, Review, Trust, Alerts, Posts, Pipeline, and Notifications. The main content area is titled 'Post Lifecycle' and includes a subtitle: 'Auto-created from confidently-negative posts. Slack and email notifications fire on entry (dry-run by default)'. A 'Refresh' button is located in the top right corner. The interface is divided into three columns:

- Ack & Reply Sent** (2 items):
 - MEDIUM (untitled) - 23h ago - r/OGPBackroom - customer service - Action needed / Close.
 - MEDIUM (untitled) - 23h ago - r/OGPBackroom - store experience - Action needed / Close.
- Actionable Items** (3 items):
 - MEDIUM (untitled) - 1d ago - r/OGPBackroom - customer service - Resolve.
 - MEDIUM (untitled) - 9d ago - r/WalmartCanada - workforce_hr - Resolve.
 - MEDIUM (untitled) - 18d ago - r/WalmartSparkDrivers - online/app - Resolve.
- Resolved** (12 items):
 - MEDIUM (untitled) - 23h ago - r/WalmartSparkDrivers - online/app.
 - LOW (untitled) - 23h ago - r/amazonprime.
 - LOW (untitled) - 23h ago - r/AmazonPrime - Rated 1 star a passive aggressive agent who was unhelpful and my amazon prime got...
 - LOW Dear Corporate - 1d ago - r/walmartgp.
 - MEDIUM (untitled) - 1d ago - r/walmartgp - customer service - How do I convince associates to sprint out the door.

Resolve modal — 2-step flow

- Step 1 · save action note + optional LLM reply
- (a) Save & open Reddit → reply copied to clipboard
- (b) OR Resolve (no-reply) → close without posting
- Step 2 · paste on Reddit → Mark Resolved
- All transitions timestamped → SLA analytics

Insights & Competitor Analysis

Strategic view — issue rankings, cross-brand pulse, LLM summaries

Priority-Negatives

Top issues ranked by
volume × severity × recency
by aspect

Competitor Pulse

Walmart vs Costco,
Target, Amazon on shared aspects

Weekly LLM Summaries

Natural-language weekly
digest + action items
+ emerging topics

Aspect Drilldown

8-aspect taxonomy
per-aspect sentiment trend
+ representative posts

Screenshot on Slide 17 (Live Demo grid) — insights_competitor page.

Notification Centre

In-app mirror of the group-routed email / Slack digest

- Every P1 / P2 email or Slack alert is mirrored in the app
- Notifications grouped by subreddit ↔ business team
- Read / unread state persisted per analyst
- Deep-links open the post directly in Review & Validate
- Powered by the same alerts rows that drive email/Slack

Priority thresholds

P1 — trust ≥ 0.70 and confidence ≥ 0.80

P2 — trust ≥ 0.50 and confidence ≥ 0.60

Source: config/models.yaml + dispatcher.py

The screenshot displays the 'Retail Sentiment INTELLIGENCE' application interface. On the left is a dark sidebar with navigation options: Brand Health, Lifecycle, Insights, Review, Trust, Alerts, Posts, Pipeline, and Notifications (which is highlighted). The main content area is titled 'Notifications' and includes a 'Reddit - Dry-run' status indicator. Below this is the 'Notification Configuration' section, which allows users to configure alerts for P1/P2 negative posts. It features a 'Sender Email' field (vishal.singh1@walmart.com) and a list of configured notification groups: 'DriverRelatedPost' (3 subreddits, 0 recipients), 'GIF_' (1 subreddit, 0 recipients), 'SAMSClub' (2 subreddits, 0 recipients), and 'W+ Membership' (1 subreddit, 1 recipient). Each group has a 'P1 + P2' priority indicator and a 'Add Group' button. At the bottom is the 'Recent Notification Log' table, which shows three entries for 'DriverRelatedPost' notifications sent via 'slack' on July 31, 04:15 PM, all of which failed.

TIME	GROUP	CHANNEL	POST	STATUS
Jul 31, 04:15 PM	DriverRelatedPost	slack	reddit_1v5jl7	FAILED
Jul 31, 04:15 PM	DriverRelatedPost	slack	reddit_1vblyrt	FAILED
Jul 31, 04:15 PM	DriverRelatedPost	slack	reddit_c_p8qqv16	FAILED

ModernBERT — Final Training Results

3-stage curriculum, 5-fold out-of-fold CV, n = 200

Overall Macro-F1

0.6272 → 0.7642

Long posts (≥ 512 tok, n = 7)

5 / 7 → 7 / 7 correct

Uplift over baseline

+13.7 pts

3-stage curriculum

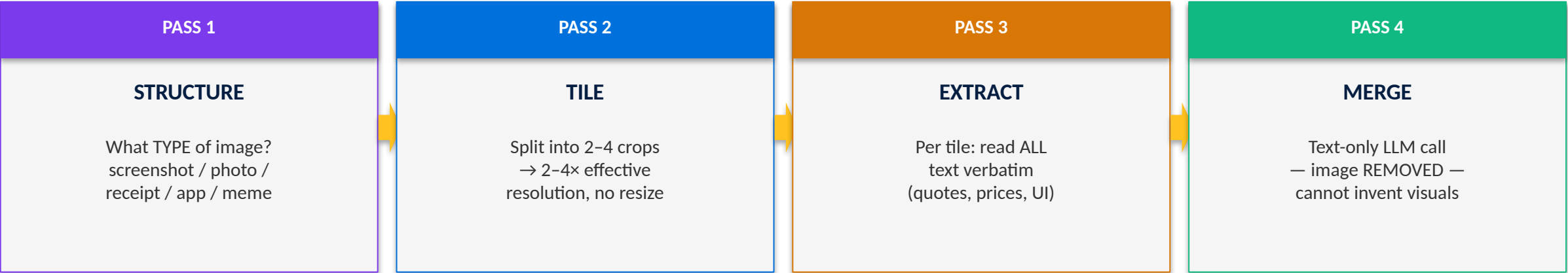
Stage	Data	Macro-F1	Δ vs baseline
Baseline (Twitter-RoBERTa)	off-the-shelf	0.6272	—
Stage 1	TweetEval sentiment (~45 k tweets)	0.6810	+5.4
Stage 2	GoEmotions-3class Reddit (~54 k, pseudo)	0.7285	+10.1
Stage 3 (final)	Walmart-200 hand-labelled	0.7642	+13.7

Per-length-bucket accuracy (5-fold OOF predictions)

Bucket (token count)	Baseline correct	ModernBERT correct	Recovered
Short-to-medium (< 512, n = 193)	138 / 193 (72 %)	159 / 193 (82 %)	+21
Long (≥ 512, n = 7, all negative)	5 / 7	7 / 7	+2

Vision Pipeline — Multi-Pass Payoff

Same model (Gemma 3 4B), same hardware, smarter calling strategy



Evaluation on 32 retail screenshots

Metric	Single-pass	Multi-pass	Change
Hallucination rate	50 %	0 %	eliminated
Text-extraction success	25 %	75 %	3× better
Retail-signal recall	40 %	81 %	2× better
Fabricated claims	many	0	eliminated
Latency / image (warm)	~5 s	~15 s	3× (accepted)

Trust-Score — End-to-End Evaluation

On the 200-post gold set, does the trust filter catch bad posts?

Low-trust share ($T < 0.4$)

15 %

Human-agreed (annotator cross-check)

12 / 15

Agreement rate

80 %

Weighted formula (decomposition shown inline in dashboard)

$$\text{trust_score} = 0.4 \times \text{metadata} + 0.3 \times \text{dedup} + 0.3 \times \text{llm_credibility}$$

Design principle — flag, don't drop

Low-trust posts are surfaced with an explanatory chip ("promotional / duplicate / thin-metadata") and can be overridden by an analyst in one click — the override is captured for the learning loop. Every constant in the formula has a stakeholder-arguable English rationale in config/models.yaml.

Evaluation Summary — Post-Midsem Numbers

All results reported against the same 200-post retail-Reddit gold set

Area	Metric	Value	Notes
Sentiment	Macro-F1 (final)	0.7642	+13.7 pts vs RoBERTa baseline
Sentiment	Long-bucket accuracy (≥ 512 tokens, n=7)	7 / 7 correct	Baseline 5 / 7 · all negative-class
Sentiment	Cross-validation	5-fold OOF	No test-set leakage
Vision	Hallucination rate	0 %	From 50 % on single-pass
Vision	Text extraction success	75 %	From 25 % on single-pass
Vision	Retail-signal recall	81 %	From 40 % on single-pass
Trust	Low-trust share (T < 0.4)	15 %	Of the 200-post set
Trust	Annotator agreement on low-trust	12 / 15	80 %
Ops	Subreddits tracked	25	Six community groups
Ops	Scheduler cadence	6 h + manual	asyncio lifespan task
Ops	Dashboard pages shipped	8	Brand Health · Alerts · Explorer · Review · Lifecycle · Insights · Pipeline · Notifications

Principal Contributions

What this dissertation delivered — end to end

C1

Offline-first RSI pipeline

Ingestion → trust → sentiment → aspects → vision → aggregation → alerts → dashboard. Zero API cost, modular, deployable to Azure.

C2

Fine-tuned ModernBERT (3-stage curriculum)

Macro-F1 0.6272 → 0.7642 overall; 0.28 → 1.00 on long posts. 5-fold OOF cross-validation, offline training.

C3

Multi-pass vision technique on Gemma 3 4B

Hallucination 50 % → 0 %, extraction 25 % → 75 %. Adapts techniques from 5 recent VLM papers without vendor-blocked models.

C4

Interpretable trust score + admission gate

trust × confidence gate flags rather than drops low-credibility posts. Every constant traceable to English rationale in config.

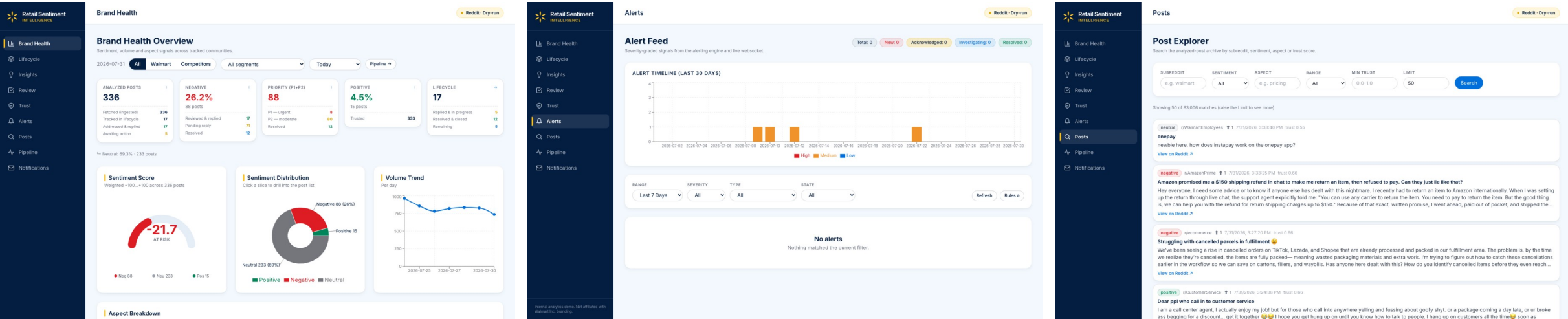
C5

HITL learning-loop dashboard

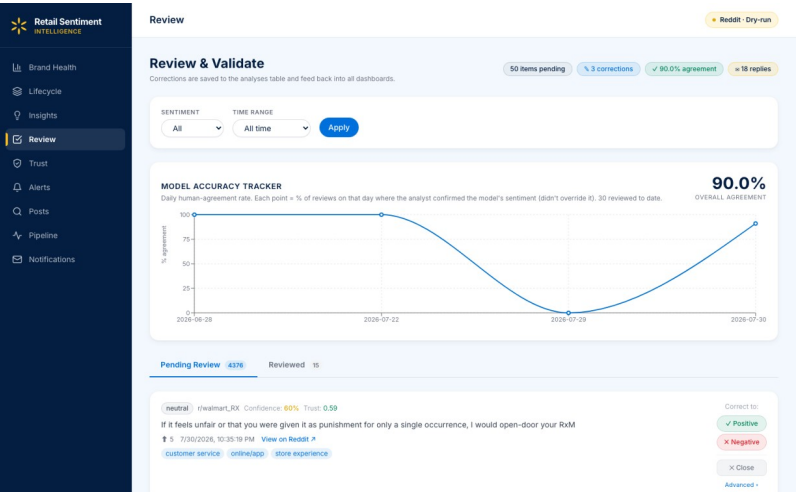
Review & Validate, Lifecycle Kanban, Insights, Notification centre. Corrections & posted replies feed few-shot reply generation and future retraining.

Live Demo — Dashboard Walkthrough

One post → alert → review → reply → lifecycle → resolved

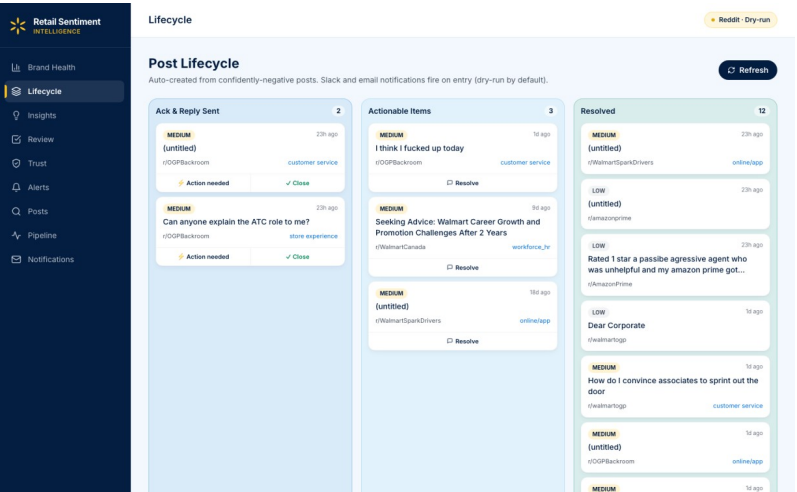


Brand Health



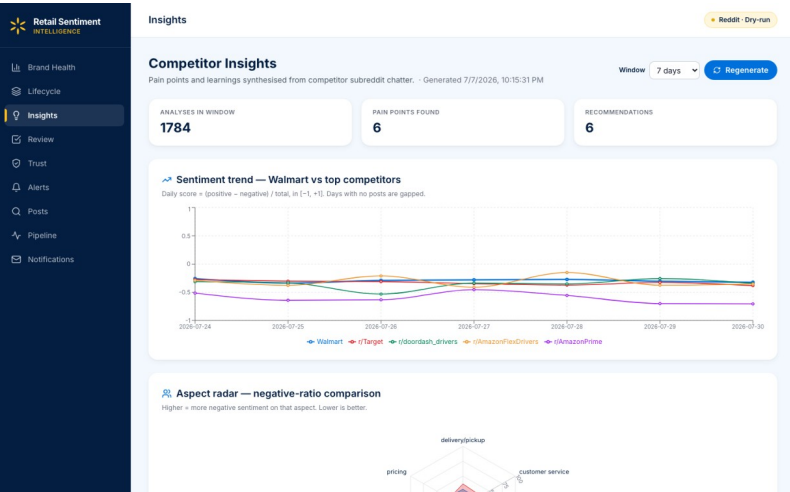
Review & Validate

Alert Feed



Lifecycle Kanban

Post Explorer



Insights

Conclusions — What Worked, What's Still Open

Honest read on 5 months of post-midsem work

What worked

- 3-stage ModernBERT curriculum landed the sentiment win we set out for (Macro-F1 0.62 → 0.76 on 5-fold OOF).
- Long-post recovery came for free once we switched off truncation (≥ 1024 -tok context): 5/7 → 7/7 correct on the ≥ 512 -token bucket.
- Removing the image on the final vision merge step ended the 50 % hallucination problem — the mechanism, not just the number, is what makes it credible.
- Trust score as flag-not-drop matched the human annotator on 12 / 15 low-trust posts. Analysts trust it because they can override it.
- HITL feedback table quietly turned into the most useful piece of infrastructure — drives few-shot on every click today, retraining tomorrow.

What's still open (honest)

- 200-post gold set is small. Numbers are OOF-CV, but a bigger blind held-out set is needed before I'd claim generalisation.
- Vision eval is 32 images — enough to sanity-check the mechanism, not enough to claim production-grade quality.
- Long bucket has only 7 posts and they are all negative-class — 7/7 correct is evidence the truncation ceiling is gone, not proof of long-text mastery.
- Reply drafts still need an analyst edit — we track edit-distance but haven't shown it dropping consistently yet.
- Everything runs on one Mac. Multi-analyst concurrency and retraining automation aren't done — they're in the roadmap on the next slide.

Future Work — Grounded 3-Horizon Roadmap

Only items with clear next steps; not a wishlist

Now → 1 month

Grow the gold set to ~500 posts using the HITL feedback the team is already producing; rerun 5-fold OOF and check the F1 delta.

1 → 3 months

Wire the monthly ModernBERT retrain into a Cron / Airflow job so it stops being a manual notebook run; add promotion gate on OOF F1.

3 → 6 months

Migrate storage to managed Postgres or Cosmos so multiple analysts can use the same feedback table concurrently; extend ingestion beyond Reddit only if the demand from the analyst team is real.

Deliberately excluded — ideas without a clear next step (auto-reply gate, seasonal P1 forecast, bilingual taxonomy) — covered in the report's Future Work chapter.

Source Code & Deliverables

Repository

https://gecgithub01.walmart.com/v0s01jh/Retail_Sentiment_Intelligence

Deliverable	Path in repo	Contents
Final Report (PDF)	docs/Sem_4/final/FINAL_REPORT_VishalSingh_2020AA05641.pdf	10 chapters + appendices + refs
Final Report (LaTeX)	docs/Sem_4/final/latex/	Reproducible xelatex source
Pipeline core	src/pipeline.py, src/ingestion/, src/analysis/	6-layer async pipeline
Sentiment training	scripts/train_modernbert_sentiment.py	3-stage curriculum runner
Evaluation notebook	evaluation/trust_score_walmart200.ipynb	Reproducible 5-fold OOF eval
Dashboard	frontend/ (React + Vite + Tailwind)	8 pages, live via WebSocket
Reproduction	Appendix B of the report · start.sh	Clone → conda → start

Thank You

Retail Sentiment Intelligence · Post Mid-Semester Presentation

Questions ?

[Repository](#)

[gecgithub01.walmart.com/v0s01jh/Retail_Sentiment_Intelligence](https://github.com/v0s01jh/Retail_Sentiment_Intelligence)

Report: docs/Sem_4/final/FINAL_REPORT_VishalSingh_2020AA05641.pdf

Vishal Singh · 2020AA05641 · BITS Pilani (WILP) · Walmart Global Tech, Bengaluru